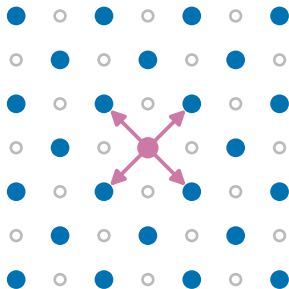
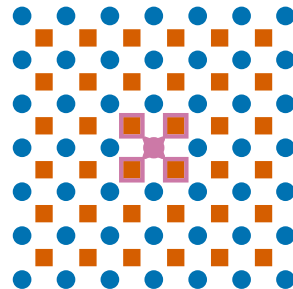


(a) checkerboard: $D_d = \{x \in \mathbb{Z}^d : \sum_{\mu} x_{\mu} \text{ even}\}$



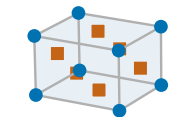
keep the filled sites
 $2d(d-1)$ neighbours at $|e|^2 = 2$

(b) body-centred: $D_d^* = \mathbb{Z}^d \cup (\mathbb{Z}^d + (\frac{1}{2}, \dots, \frac{1}{2}))$

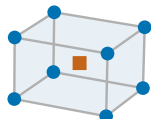


integer sites plus cell centres
 2^d diagonals at $|e|^2 = d/4$

(c) $d=3$: the dual is a different lattice



$D_3 = \text{fcc}, \tau = 12$



$D_3^* = \text{bcc}, \tau = 8$

(d) how the dual behaves with dimension

